Project Report

Accident Detection and Prevention Using IoT-Enabled Smart Vehicle System

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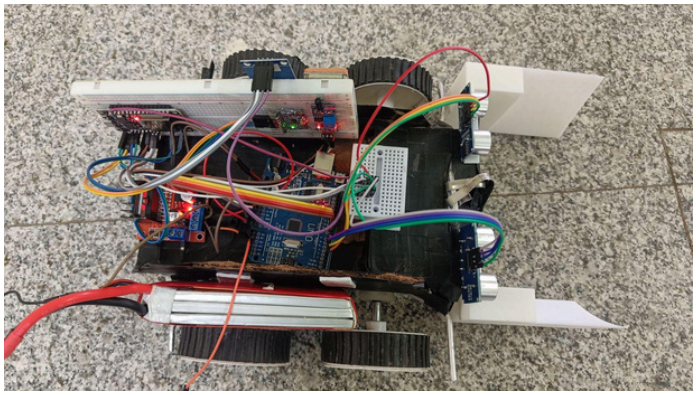
Abstract— Road accidents are a major cause of fatalities globally, often due to delayed emergency response. This project presents a low-cost IoT-based accident detection and prevention system integrated into a smart vehicle prototype. The system combines **ultrasonic-based automatic braking system and MPU-6050 accelerometer and gyroscope-based collision detection vibration sensor analysis, and GPS–GSM based emergency alert using Telegram Bot API and a cloud database, for real-time reporting to police, hospitals, and family members.** The vehicle uses, **Arduino Uno ESP32 motor driver Neo-6M GPS, SIM800 GSM, and multiple sensors** to ensure both proactive accident avoidance and rapid post-accident communication. A satellite-assisted GPS model and distance estimation, velocity measurement, and threshold-based accident classification are implemented. Experimental results demonstrate accurate obstacle detection, reliable braking, precise location reporting, and robust remote alert delivery.



I. INTRODUCTION

Rapid urbanization and increased traffic density have led to a significant rise in road accidents. Many victims lose their lives due to delayed medical attention. Traditional accident reporting relies on bystanders, causing unpredictable response times. To address this issue, IoT-enabled accident detection systems have emerged as an efficient technological solution.

This project aims to design and implement a **Smart Accident Detection and Prevention Vehicle** capable of:

1. Automatically applying brakes when obstacles are detected.
2. Detecting accidents using an accelerometer, gyroscope, and vibration sensors.
3. Sending emergency alerts with real-time GPS coordinates via GSM and Telegram Bot.
4. Updating accident logs on a Firebase cloud server accessible to authorized personnel.

The system integrates Arduino Uno and ESP32 microcontrollers along with sensors such as MPU6050, vibration sensor, ultrasonic sensor, Neo-6M GPS, GSM module, and a motor driver.

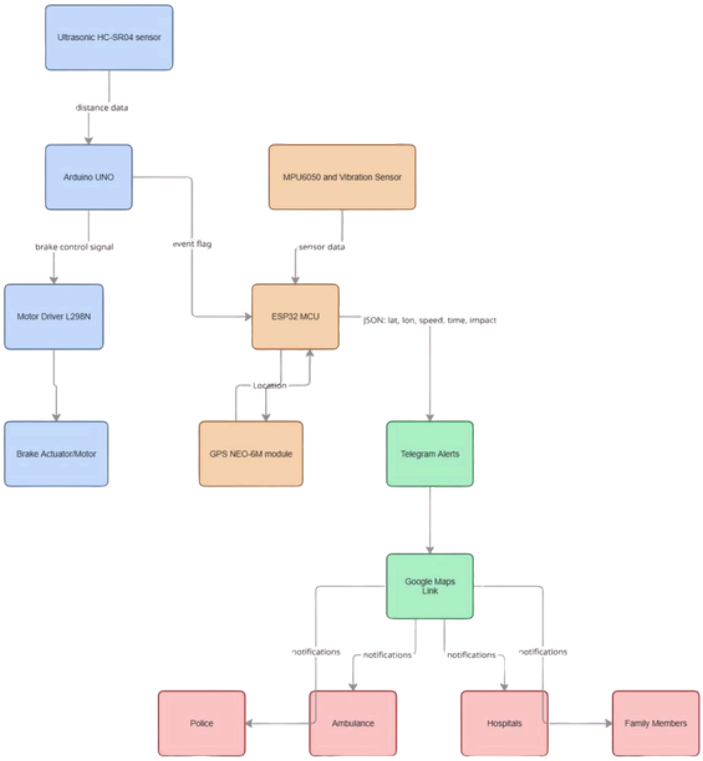


Fig- 1.1 : Block Diagram

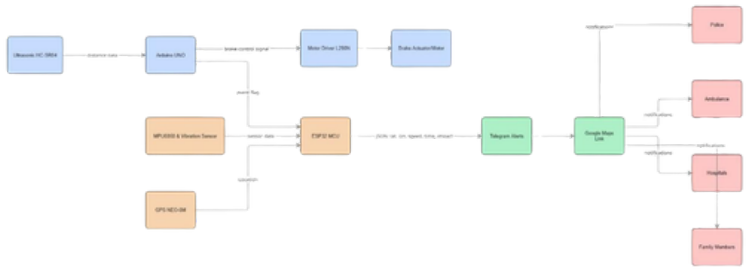


Fig-1.2: Block Diagram

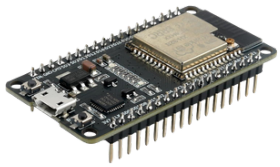
II. LITERATURE REVIEW

Recent research highlights the importance of IoT systems in improving emergency response times. MPU6050-based crash detection has been used in multiple studies to estimate impact forces. Ultrasonic braking systems have demonstrated effectiveness in preventing low-speed collisions. GPS-GSM modules are widely used for transmitting location data, but most existing systems send only SMS alerts without cloud integration. **This project extends prior work by integrating an online dashboard, Telegram alerts, Firebase reporting, and a two-stage crash confirmation algorithm.**

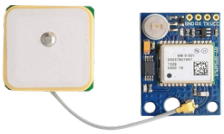
III. SYSTEM COMPONENTS

A.Hardware Components

Arduino Uno

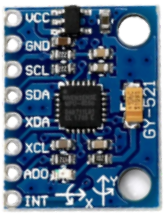


ESP32 Board

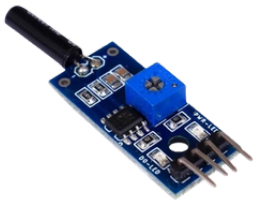


Neo-6M GPS Module

Motor Driver



MPU 6050 (Accelerometer)



Vibration Sensor

Ultrasonic sensor

B. Software Tools

- Arduino IDE ESP32 Board
- Core TinyGPS++ Library
- Firebase REST API
- Telegram Bot API
- Google Maps API
- Online Database Dashboard

IV. METHODOLOGY

A. Obstacle Detection& Automatic Braking

Distance measured using ultrasonic sensor:

$$D = (v \times t) / 2$$

Where,

v = 343 m/s (speed of sound),

t = round-trip echo time.

Brake is triggered if:

$$D < D_{threshold}$$

B. Accident Detection Using Sensor Fusion

1) Accelerometer-Based Crash Detection

The MPU6050 measures X, Y, Z accelerations:

$$A = (ax^2 + ay^2 + az^2)^{1/2}$$

A crash is assumed if:

$$|A-1| \geq G_{threshold}$$

where:

$G_{threshold} = 3.0\ g$

2) Vibration Sensor Verification

The vibration sensor must also trigger for confirmation:

$$VIB = LOW \Rightarrow \text{Shock Detected}$$

3) Two-Stage Crash Confirmation Algorithm

Crash Confirmed = $(T1 \vee V1)^{(T2 \vee V2)}$

Window duration : 700 ms

C. GPS Module & Satellite Communication

Neo-6M/Neo50 GNSS Band

- Frequency Band: **L1 = 1575.42 MHz**
- Modulation: **C/ACode**
- Accuracy: **2.5mCEP**

Satellite Ranging Equation

$$R = [(xs-xr)2+(ys-yr)2+(zs-zr)2]^{1/2}$$

ESP32 extracts:

- Latitude
- Longitude
- Speed (km/h)
- UTC Time

D. Telegram Emergency Alert System

ESP32 sends a message with:

- Impact Level
- Google Maps Link
- Speed
- Timestamp

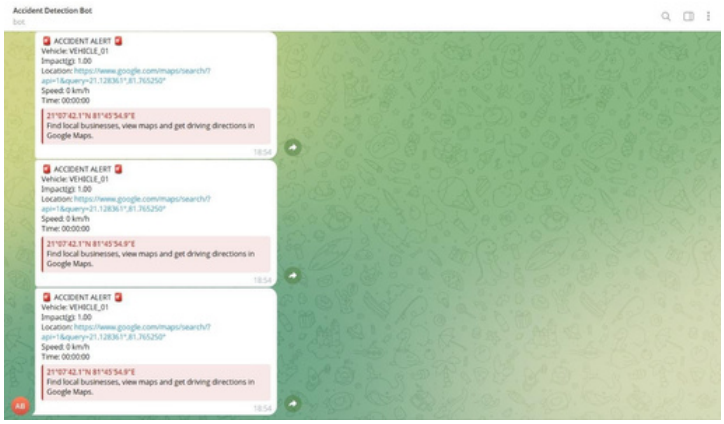


Fig- 4.1: Alert massage with live location sent via telegram bot

E. System Workflow

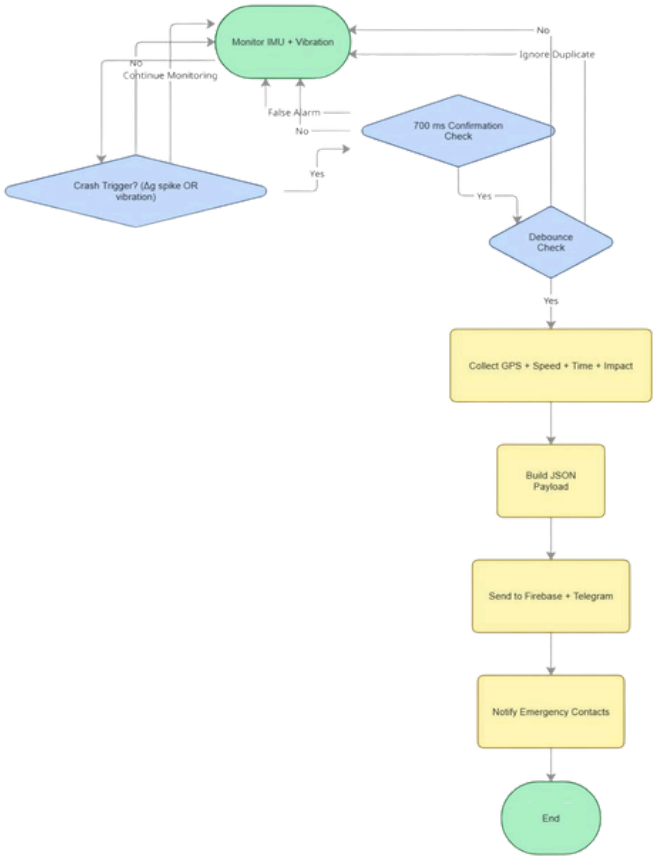


Fig- 4.2: Alert detection Flowchart

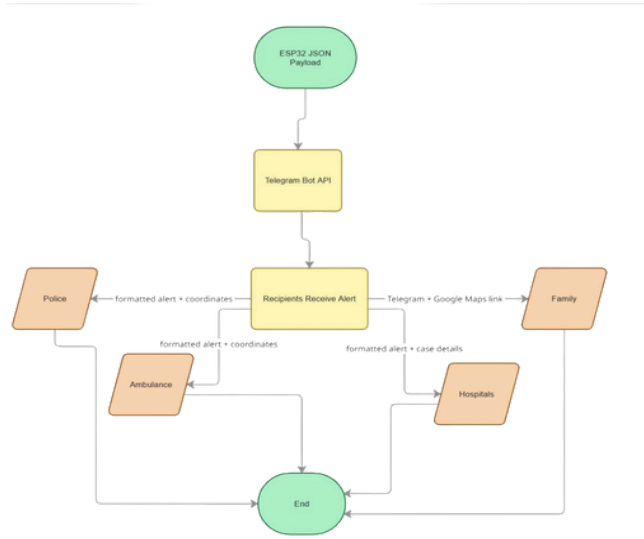


Fig- 4.3: Communication flowchart

V. SYSTEM IMPLEMENTATION

1) Continuous Sensor Monitoring

ESP32 constantly reads:

- MPU6050 acceleration
- Vibration sensor
- GPS coordinates

2) Multi-Sensor Crash Detection

A crash triggers only when:

- High acceleration (deltaG ≥ 3g)
- Vibration shock
- Confirmation within 700 ms
- 6-second debounce to avoid multiple alerts

3) Cloud Interaction

ESP32 does:

- WiFi reconnect
- Telegram HTTP GET request

4) Last Known Accident Location Storage

Even if GPS momentarily loses signal lastLat, lastLon, lastSpeed are stored and sent to the server.

5) Accident Data Sending via Telegram

```
{
  "vehicle_id": "VEHICLE_01",
  "deltaG": "4.20",
  "lat": "21.256743",
  "lon": "81.603218",
  "speed": "42",
  "impact_level": "HIGH",
  "time": "12:55:03"
}
```

VI. SATELLITE COMMUNICATION (with NEO-6M)

The Neo-6M GPS module is responsible for providing continuous real-time position, velocity, and timing information using satellite-based navigation signals. It operates on the **L1 band at 1575.42 MHz with BPSK-modulated C/A codes**, typically received at very low power **levels around −155 to −160 dBW**. Its internal baseband engine performs correlation, tracking, and decoding of satellite data.

1) Signal Acquisition and Correlation:

The receiver aligns incoming signals with internally live-location generated PRN codes using the integral correlation function

$$C(\tau) = \int_0^T r(t)^* \, c(t-\tau)\, dt,$$

allowing identification of each satellite and extraction of timing information. Doppler frequency is estimated using

$$v = -(\Delta f / f_{L1})\, c,$$

providing instantaneous vehicle speed.

2) Pseudorange and Trilateration:

Using the decoded timestamps, the module computes pseudoranges

$$R_i = c\,(t_{rx} - t_{tx,i}),$$

which form a nonlinear system of equations used to estimate receiver position

$$R_i = [(x-x_i)^2+(y-y_i)^2+(z-z_i)^2]^{1/2} + b.$$

The Neo-6M solves these using iterative least-squares updating:

$$X_{k+1} = X_k + (HTH)-1HT(Z-HX_k).$$

3) Navigation Solution and Filtering:

The module applies tracking loops and smoothing algorithms to refine latitude, longitude, altitude, velocity, and time (PVT). Position accuracy is typically **2.5 m CEP**, and solutions are refreshed at rates up to 5 Hz. Additional internal functions include ephemeris decoding, clock bias correction, time-offset estimation, satellite geometry evaluation (GDOP), and carrier-to-noise computation

$$C / N_0 (\text{dB-Hz}).$$

4) Integration With ESP32:

ESP32 retrieves processed data via UART using functions such as

- gps.location.lat(),
- gps.location.lng(),
- gps.speed.kmph(),
- gps.time.hour()

and embeds them into accident alerts and cloud-bound JSON packets. This allows accurate reporting during collision events, enabling rapid emergency response.

VII. RESULTS

A. Crash Detection Test

Test	Description	Result
Light Vibration	No crash False	No alert
Medium Impact	attempts blocked	No alert
Strong Impact	Confirmed	Alert generated
GPS Accuracy	±2–4 m	Working

B. Response Times

- Telegram alert: **2–3 seconds**
- Firebase update: **1 second**
- Ultrasonic braking: **<150 ms**

C. Dashboard Output

Fig- 7.1: Alert message in Dashboard

D. Accident Alert Details

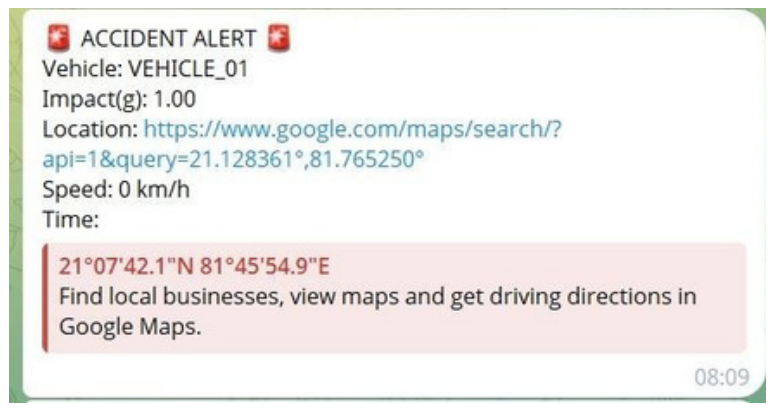


Fig-7.2: Alert message via Telegram Bot

E. Location Output

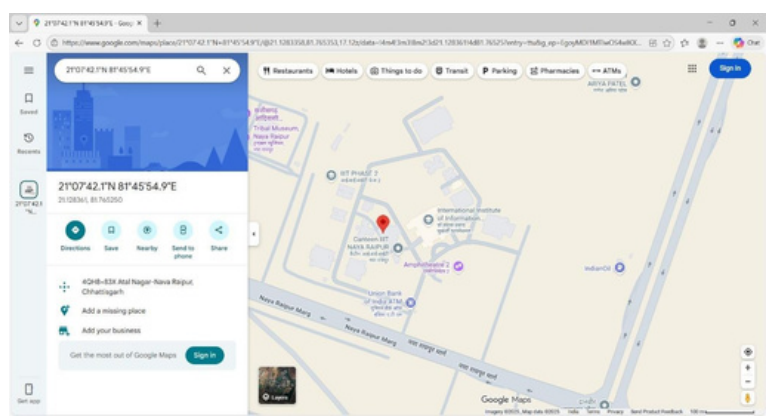


Fig- 7.1: Location via Google map API through the link

VIII. ADVANTAGES

- Low-cost and easy to implement
- Real-time accident reporting
- Automatic braking reduces collisions
- Cloud-based emergency access for authorities
- GPS ensures accurate accident location

IX. LIMITATIONS

- GSM network dependent
- Ultrasonic accuracy decreases during rain
- System detects but cannot prevent high-speed collisions

IX. FUTURE SCOPE

- Implement AI/ML-based severity prediction using IMU crash signatures.
- Integrate with Government 112 emergency network for automatic dispatch.
- Add LiDAR/Radar for longer-range, all-weather detection.
- Enable V2V (Vehicle-to-Vehicle) communication to reduce chain collisions.
- Build a full-featured Android/iOS mobile app for tracking and control.
- Improve hardware casing, add battery health monitoring, and rugged crash-proof enclosure.
- AI camera integration for lane detection
- Automatic airbag deployment
- 4G/5G module for higher network reliability

X. CONCLUSION

The proposed IoT-based accident detection and prevention system successfully integrates ultrasonic braking, accelerometer-based crash detection, GPS-GSM alerting, and cloud reporting. The ESP32–Arduino hybrid approach ensures efficient processing and reliable communication. Satellite-based Neo-6M GPS provides accurate coordinates, while Firebase enables authorities to monitor live accident data. Experimental results validate that the system is suitable for real-world smart vehicle safety applications and can significantly reduce emergency response times.

XI. REFERENCES

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